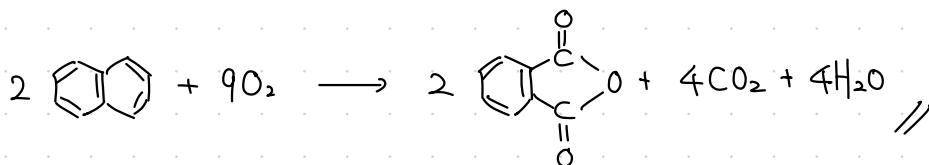
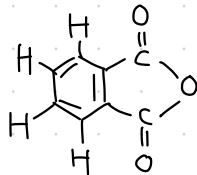
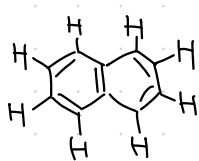
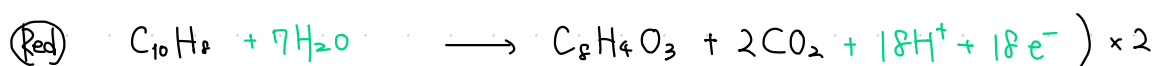
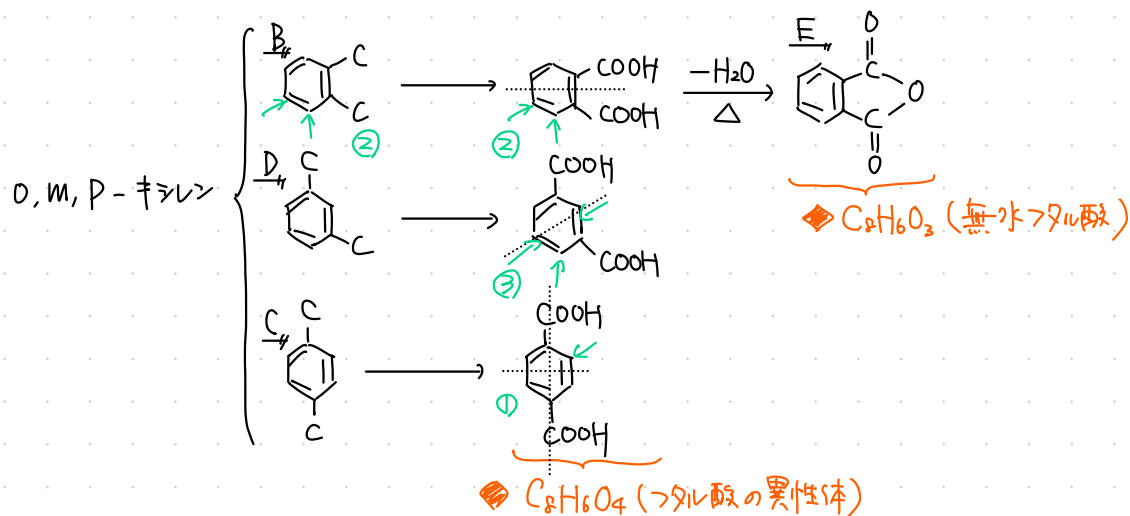
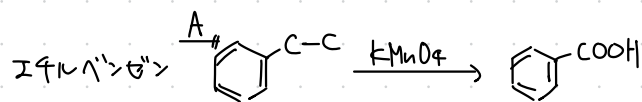


4-① C_8H_{10} $U = 4$ $\text{C}_6\text{H}_5 \times 1$

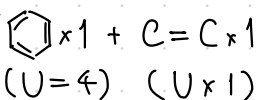


(酸化だから、余った $\text{C} \xrightarrow{\text{O}_2} \text{CO}_2$,
 $\text{H} \xrightarrow{\text{O}_2} \text{H}_2\text{O}$ になると考え、
 係数合わせでも良い)

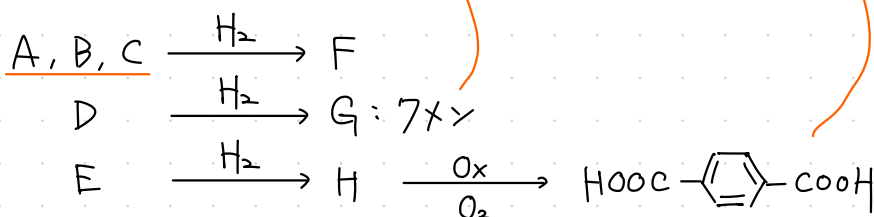
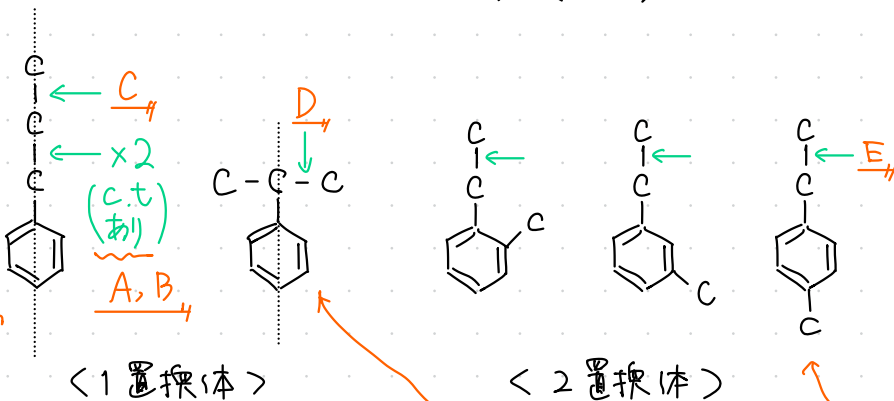
4-2

C_9H_{10}

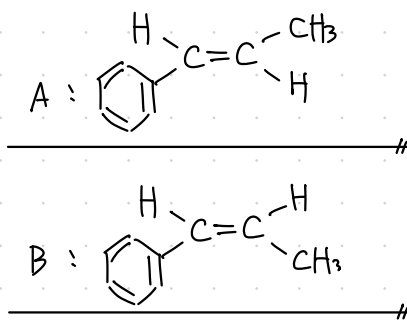
$U = 5$



内2.

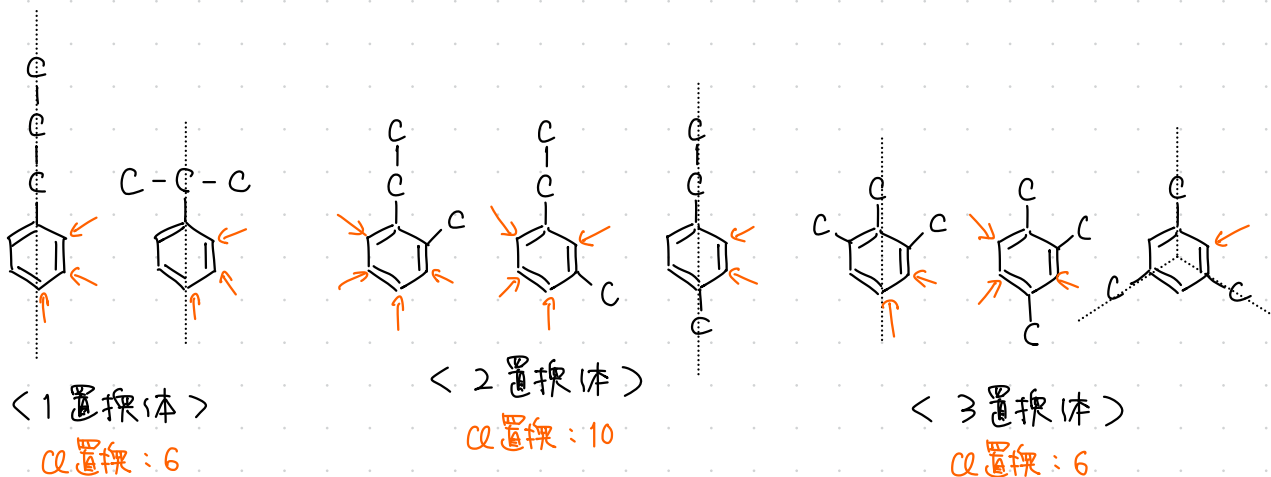


A, B: シス・トランス異性体
 融点 $A > B$
 トランス シス



内1. 8

内3. 22



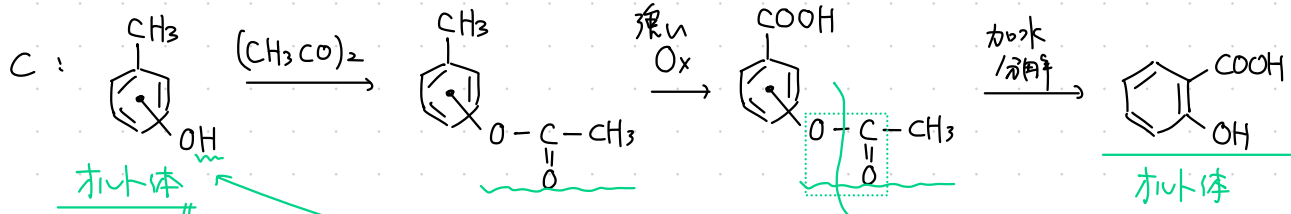
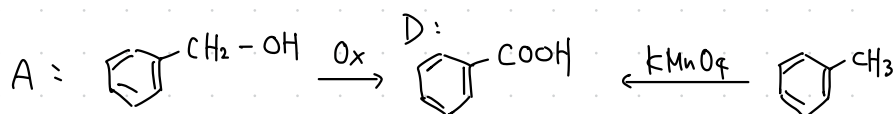
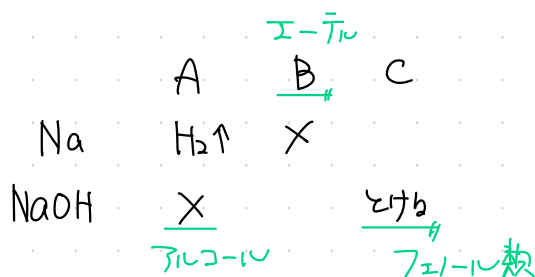
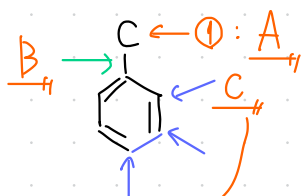
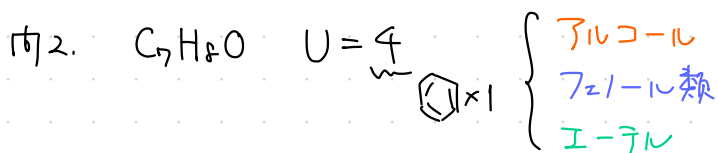
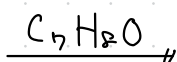
4-③

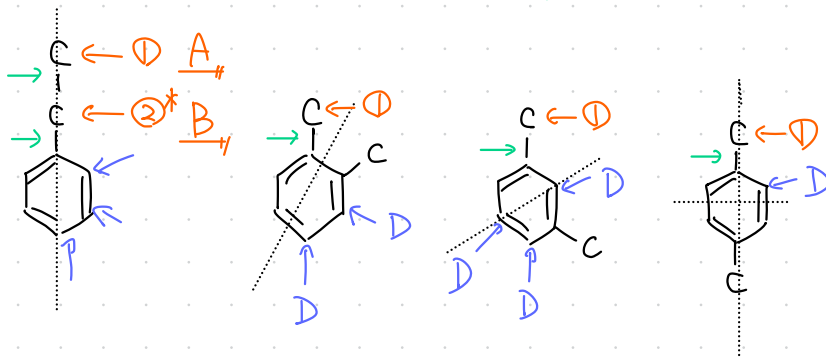
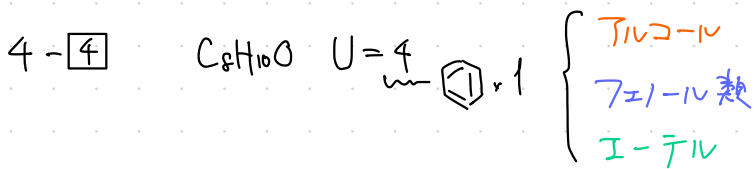


条件より $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

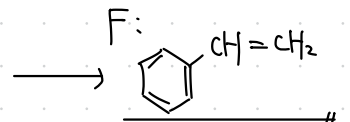
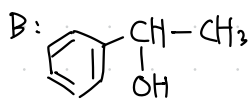
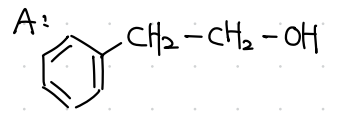
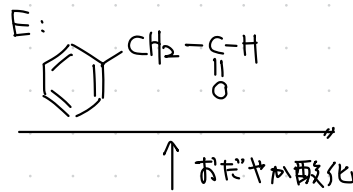
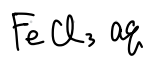
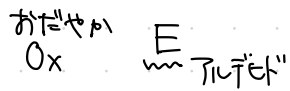
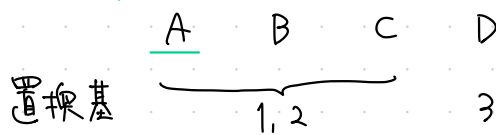
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

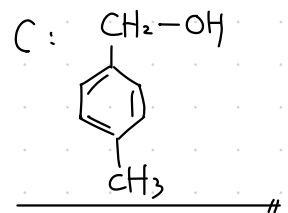
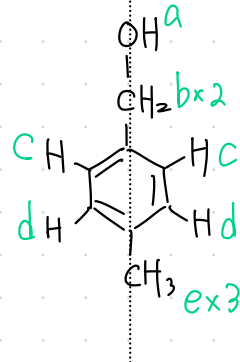
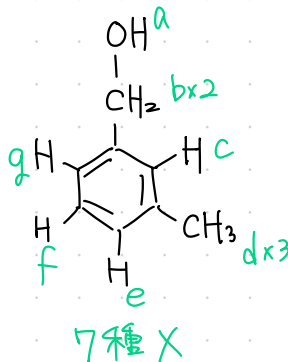
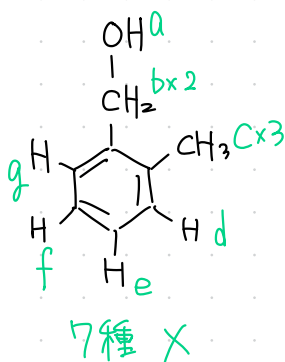




第1級
アルコール



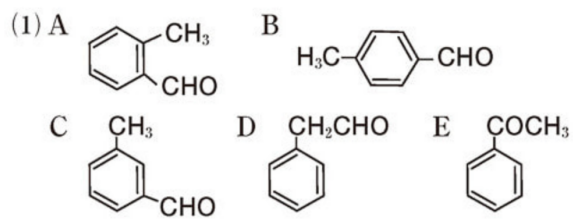
C: 1 or 2 置換体のアルコール



内2

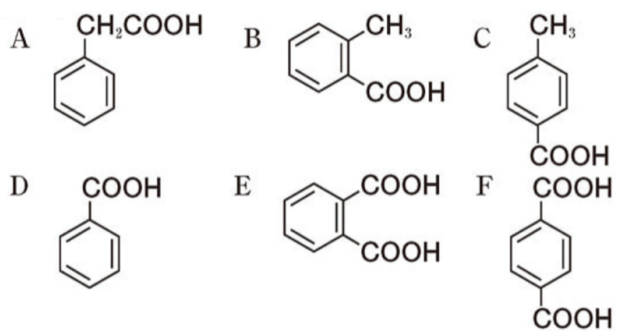
D: フェニル類 (3 置換体), $\underline{6}$

4 - 5

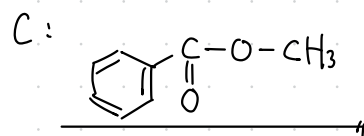
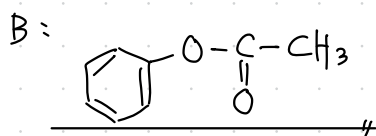
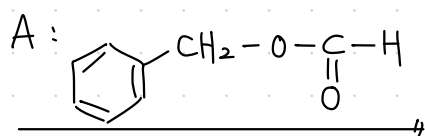
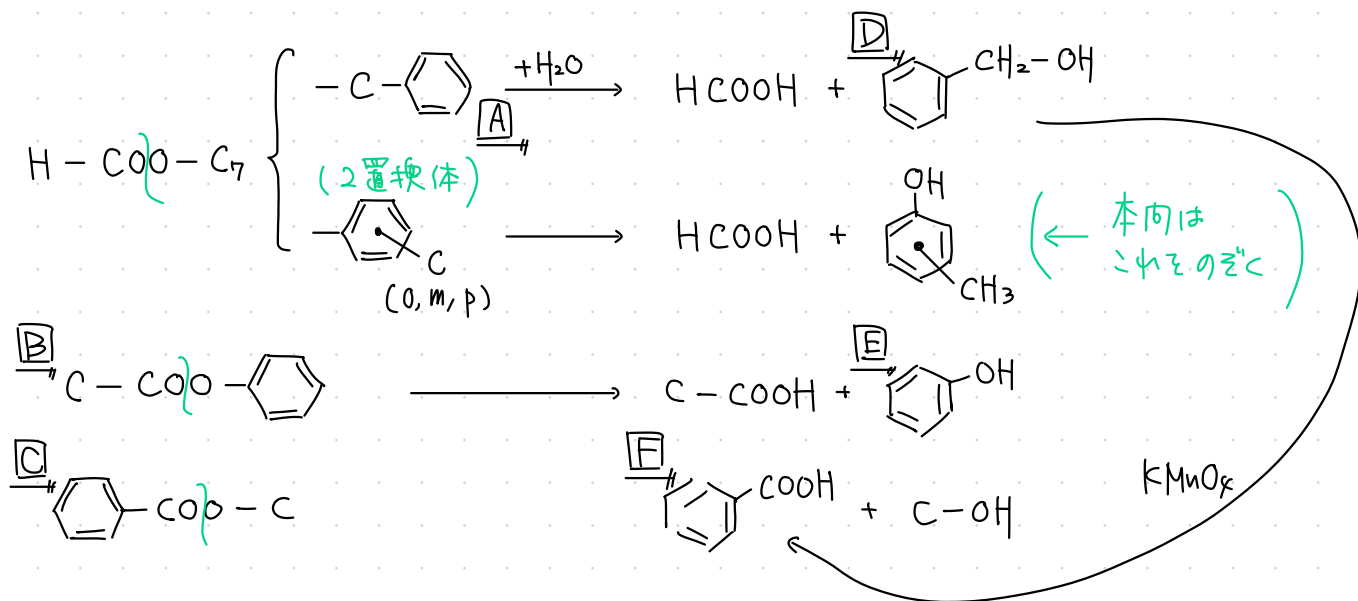
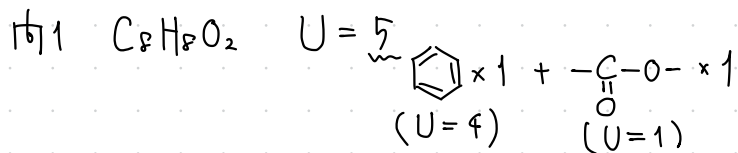


- (2) Gは*p*-置換体であるため、*o*-, *m*-置換体よりも分子の形が対称的である。したがって、結晶化しやすいため、融点は最も高くなる。

4 - 6

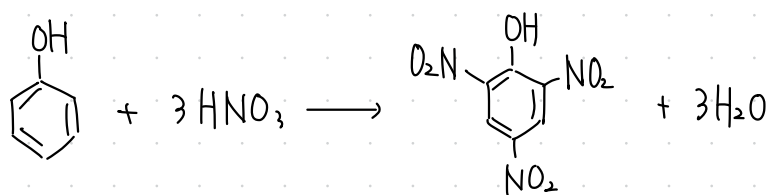


4-7



例2. E

例3



トリニトロフェノール
 $(2, 4, 6-\text{ト})$ -ニトロフェノール

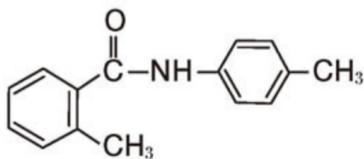
4-8

(1) $C_{15}H_{15}NO$

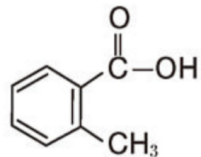
(2)

$$U = \frac{2 \times 15 + 2 + 1 - 15}{2} = 9$$

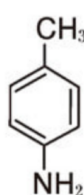
A



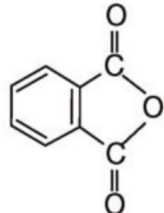
B



C



D



$C_{15}H_{15}NO$

$$U = \frac{2 \times 15 + 2 + 1 - 15}{2} = 9 = \begin{matrix} \text{アミド} \\ -\text{C}(=\text{O})-\text{NH}- \end{matrix} + \text{ベンゼン} \times 2$$

アミド
A

加水分解
 H^+

(I)

カルボン酸
B
木片体

C_8 以上

$KMnO_4$

B'

$-H_2O$

Δ

酸無水物D

木片体

(II)

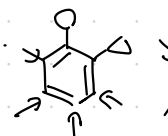
アミンCのHCl塩

OH^-

アミンC

Wittig

2置換体



$-NH_2$,
 $-CH_3$ あり

アミド A

カルボン酸 B

アミン C

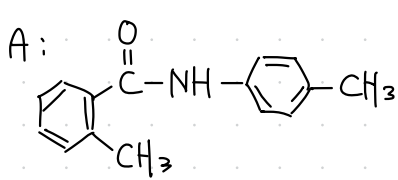
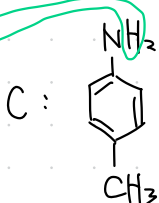
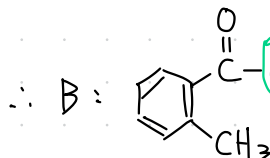
C_{15}

C_8 以上

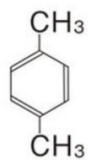
C_7 以上

C_8

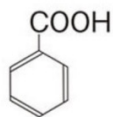
C_7



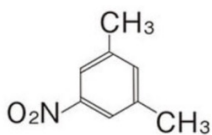
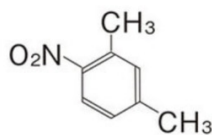
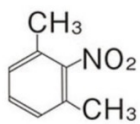
問 1 化合物 D :



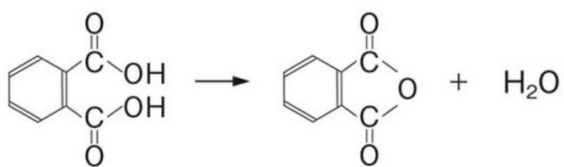
問 2 化合物 E :



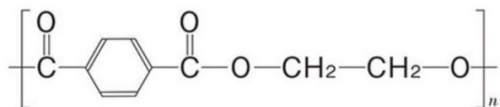
問 3



問 4



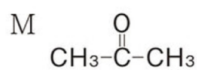
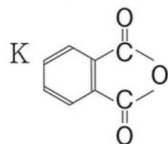
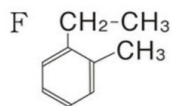
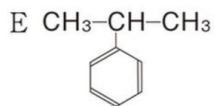
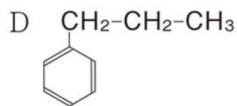
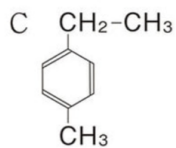
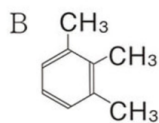
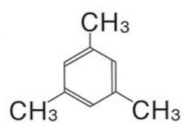
問 5



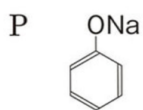
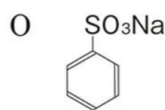
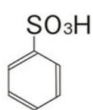
4 - 10

問1 (ア) エチレングリコール(1,2エタンジオール) (イ) 酸素
(ウ) ヨウ素

問2 A



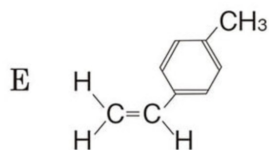
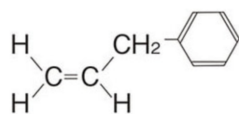
問3 N



4 - 11

問1 2個 問2 クメン

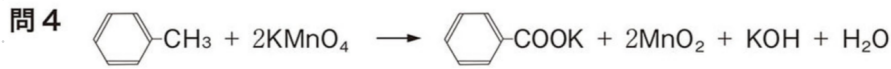
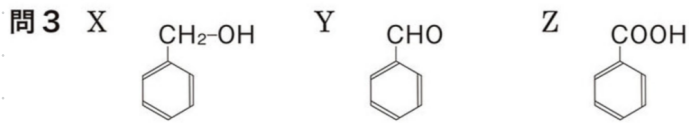
問3A



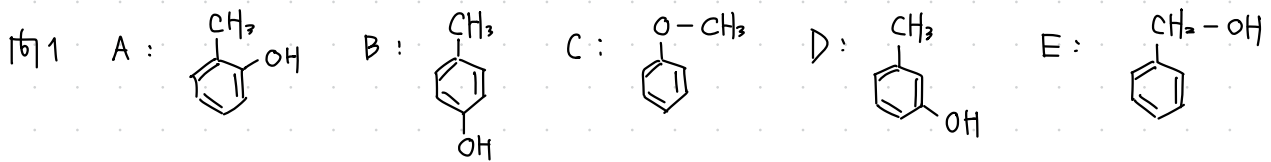
4-12

問1 気体…水素 官能基…ヒドロキシ基

問2 試薬… FeCl_3 色…紫



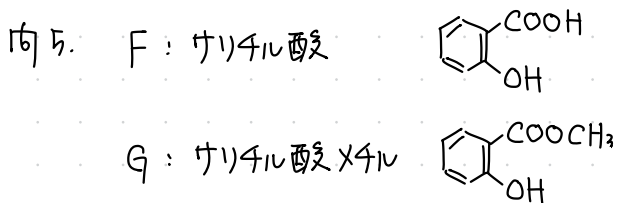
4-13



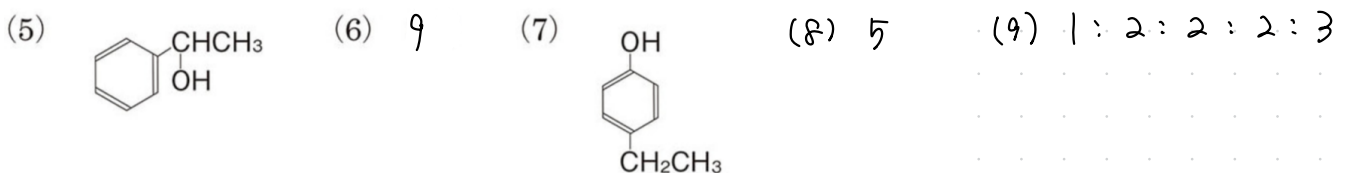
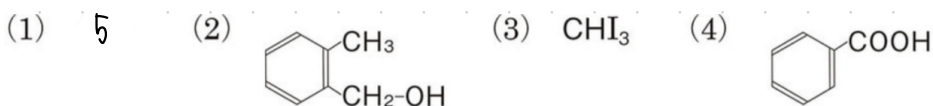
問2 NaOH aq を加えると A, B, D は 中和して均一な溶液になる (とける)。
(C, E は 2層に分離する) ↑ これは間違い。見ためや1においでわかることをかく。

問3 ヒドロキシ基 を持たないため、水素結合 ができず、分子間力が弱いから。

問4 金属 Na を加えたとき、C は H_2 を発生しない。

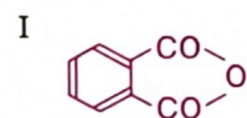
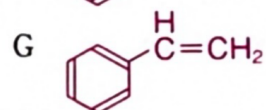
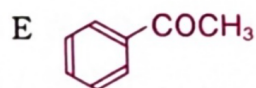
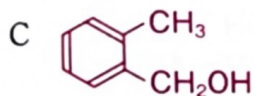
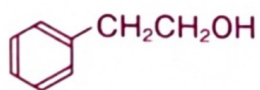


4-14

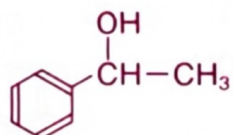


4 - 15

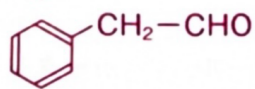
問1 A



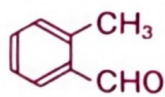
B



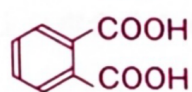
D



F



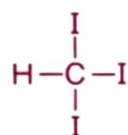
H



問2 B

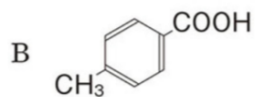
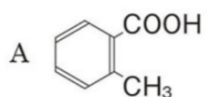
問3 ポリスチレン

問4 記号 B 構造式



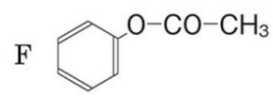
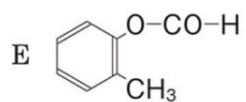
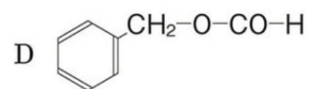
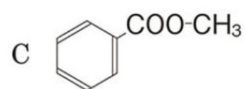
4 - 16

問1 (1)



(2) ナフタレン, o-キシレン

問2 (1)

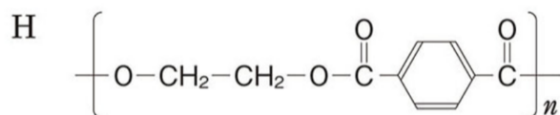
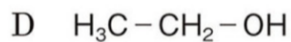
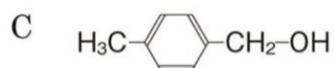
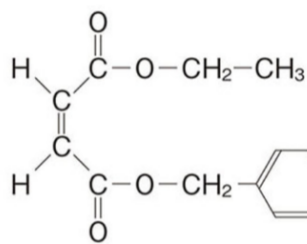


(2) 2,4,6-トリブロモフェノール

4 - 17

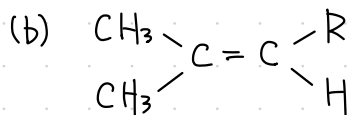
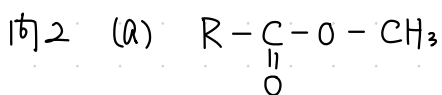
問1 $C_{14}H_{16}O_4$

問2 A

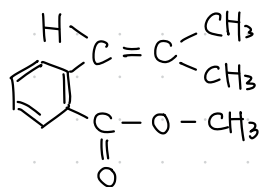


4 - 18

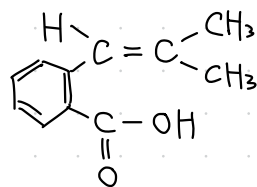
問1 $C_{12}H_{14}O_2$



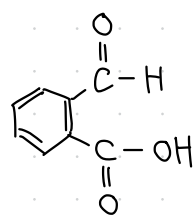
問3 A



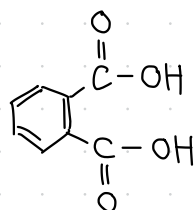
B



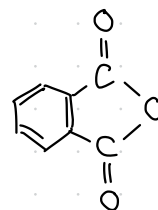
C



D

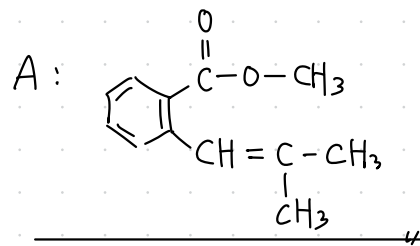
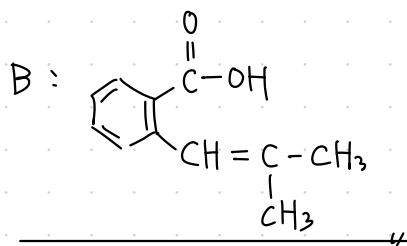
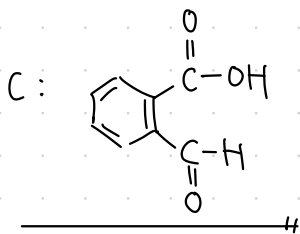
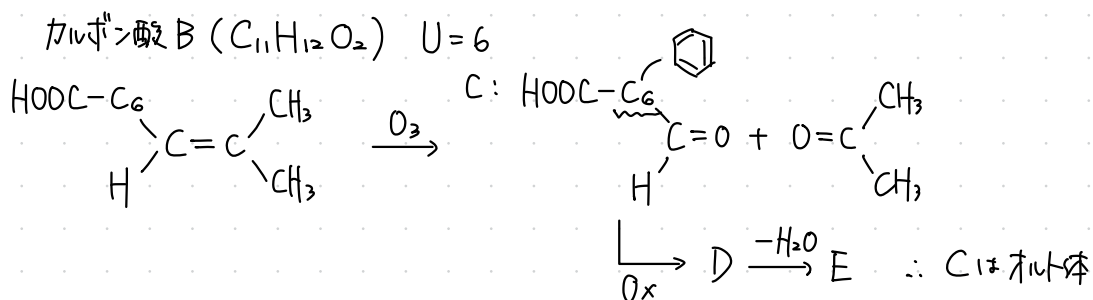
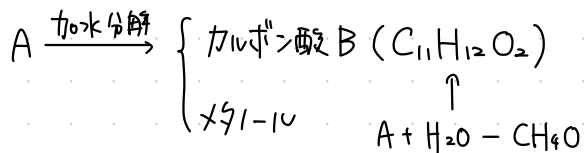
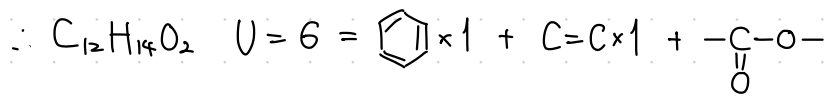


E



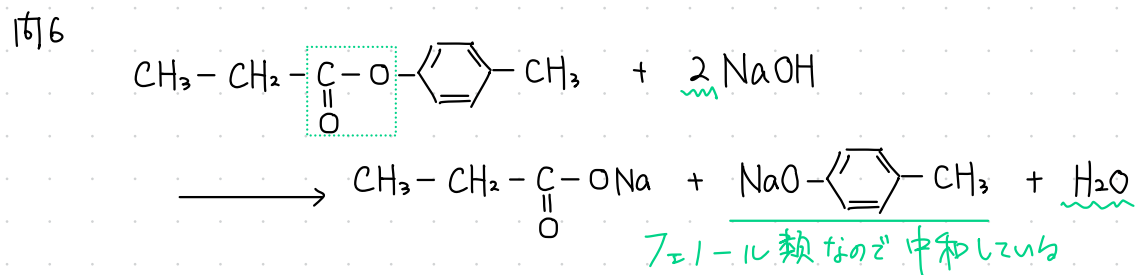
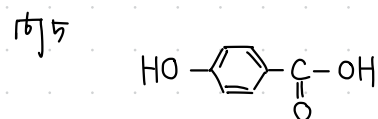
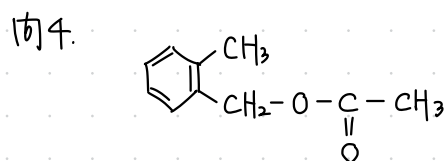
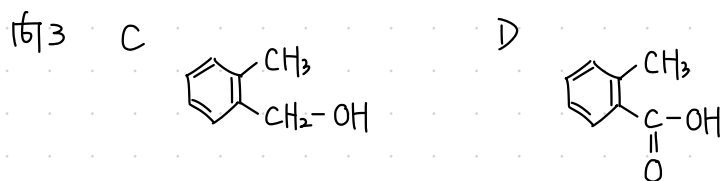
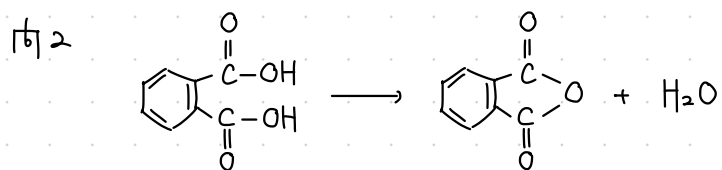
問1 $C_xH_yO_z$ を求める,

$$x = \frac{190 \text{ g} \times \frac{75.8}{100}}{12 \text{ g/mol}} = 12, \quad y = \frac{190 \text{ g} \times \frac{7.4}{100}}{1.0 \text{ g/mol}} = 14, \quad z = \frac{190 \text{ g} \times \frac{16.8}{100}}{16 \text{ g/mol}} = 2$$

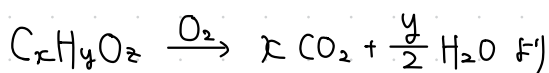


4 - 19

問1 あ C_5H_6O n 164 う $C_{10}H_{12}O_2$



問1 $\Delta f = k_f m$ $2.56 = 5.12 \times \frac{410 \times 10^{-3} \text{ g}}{M \text{ g/mol}} \times \frac{1}{5 \times 10^{-3} \text{ kg}}$ $\therefore M = 164 \text{ (u)}$



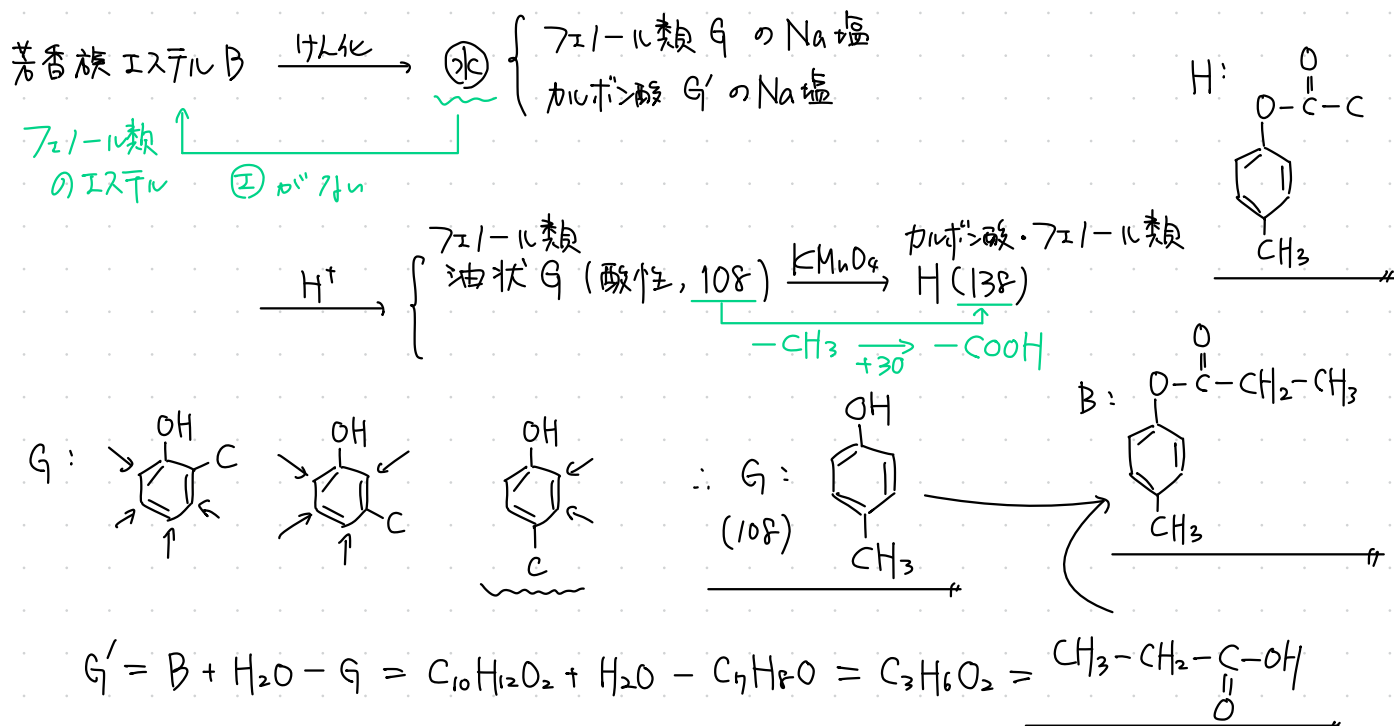
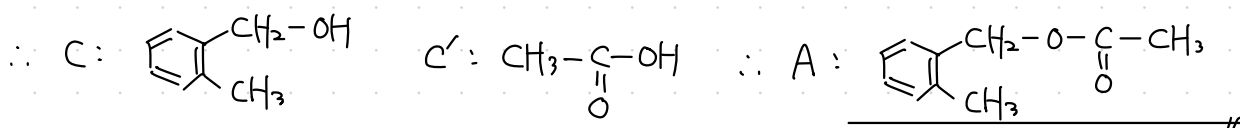
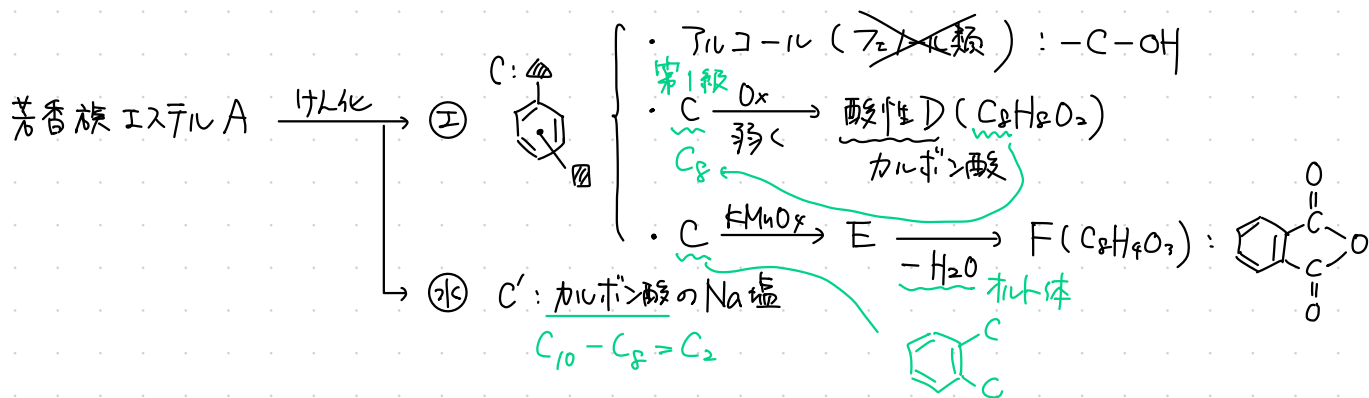
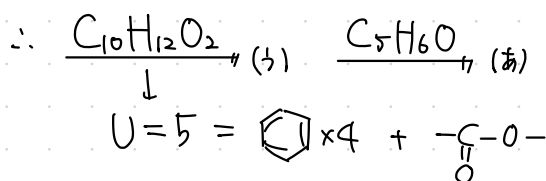
$C_x H_y O_z : CO_2 = 1 : x$ したがって,

$\frac{8.2 \text{ g}}{164 \text{ g/mol}} \times x = \frac{22 \text{ g}}{44 \text{ g/mol}} \therefore x = 10$

$C_x H_y O_z : H_2O = 1 : \frac{y}{2}$ したがって,

$\frac{8.2 \text{ g}}{164 \text{ g/mol}} \times \frac{y}{2} = \frac{5.4 \text{ g}}{18 \text{ g/mol}} \therefore y = 12$

$M = 12x + y + 16z = 120 + 12 + 16z = 164 \therefore z = 2$

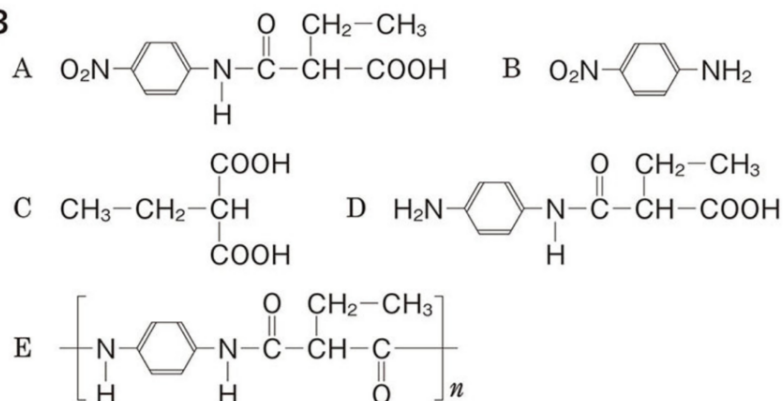


4- 20

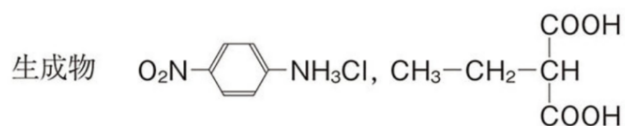
問1 C 原子…52.4 % H 原子…4.8 % N 原子…11.1 %

問2 $\text{RCOOH} + \text{NaHCO}_3 \rightarrow \text{RCOONa} + \text{CO}_2 + \text{H}_2\text{O}$

問3

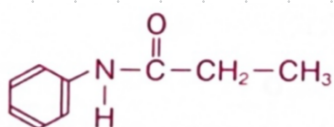


問4 塩酸を用いることで化合物 A のアミド結合が加水分解されてしまうため。

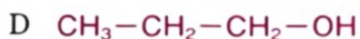
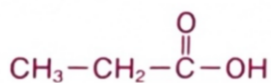


4- 21

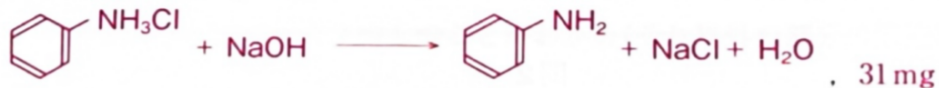
問1 A



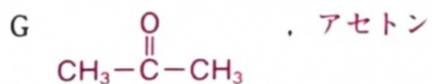
C



問2

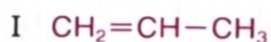
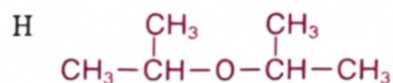


問3 F $\text{CH}_3\text{-CH}_2\text{-O-CH}_3$, エチルメチルエーテル



問4 F, E, D

問5



問1 $C_{19}H_{20}O_4$

